

SIGNOVA '26 – A National Level Technical Symposium

EVENT RULES BOOK

TECHNICAL EVENTS

1. Paper Verse

Innovate. Present. Inspire

Showcase your ideas and explore the power of technology through impactful paper presentation.

Team size: Individual or team (2-3 members)

Presentation Guidelines:

- Topics must be technical, innovative, or research-oriented.
- The work presented should be original. Plagiarism is strictly prohibited.
- Presentation must be in PPT or PPTX or PDF format.

Time Limit:

Presentation: 5 minutes

Q&A: 2 minutes

Domains:

- Semiconductors & Nano Technologies
- Communication & Networks
- Robotics & Automation
- Analog & Digital electronics
- Autonomous Vehicles and ADAS
- Biomedical Instrumentation
- Embedded systems & IOT
- AI/ML
- Open Innovation (Own a stage)

General Rules:

- Report to the venue 30 minutes before the event
- Formal attire with valid college ID.
- Mobile phones should remain silent during presentations.
- Exceeding the time limit may lead to deduction of marks.
- The decision of the judge is final and binding.

2. Bid and Build

Decode. Bid. Build

A strategic electronics design challenge where participants analyze problem statements, manage virtual resources, and construct functional systems through simulation. Success depends on technical knowledge, budgeting skills, and efficient design decisions.

Team size: Team (2-4 members)

Number of Rounds: 2 (each round 3 phases)

General Rules:

- Each team may consist of 2–4 members.
- Only the components acquired during the auction may be used. Incorrect or unnecessary component purchases may result in penalty points if returned.
- Teams are allowed to use AI for connections and code generation, but must complete all wiring, integration, and debugging themselves.
- The final score will be based on: System functionality, Accuracy of implementation, Remaining Signova Coins, Completion time and Circuit optimization.
- Any unfair practices will result in disqualification. Judges' decisions will be final and binding.

ROUND 1 – FOUNDATION BUILD

Phase 1: Problem Analysis

Teams will be provided with one or two basic problem statements.

Examples:

Automatic street light using an LDR. Temperature monitoring system with LCD display.

Teams will have 2 minutes to analyse the requirements and identify the components needed.

Phase 2: Component Auction Using their allotted Signova Coins.

Teams must bid for the virtual electronic components required to solve the problem.

Participants should spend wisely, as unused coins contribute to their final score.

Phase 3: Design & Simulation

Teams must design, wire, program, and simulate the required circuit using an approved online simulator (e.g., Wokwi or Tinkercad).

The completed circuit must successfully demonstrate the expected output.

ROUND 2 – INTEGRATED SYSTEM CHALLENGE

Phase 1: Advanced Problem Analysis

- Teams will be given three interconnected problem statements that must be combined into a single working system.

Example Challenge:

Problem 1: Detect motion using a PIR sensor.

Problem 2: Display alert messages on an LCD display.

Problem 3: Trigger an automatic buzzer and LED warning system when motion is detected.

Final Objective: Integrate all three functionalities into one complete security monitoring system.

- Teams will have 2 minutes to analyze the challenge and identify the required components.

Phase 2: Component Auction

- Teams must bid for all necessary components using their remaining Signova Coins.
- Careful budgeting is essential, as incorrect purchases may affect both performance and scoring.

Phase 3: System Design & Simulation

- Teams must: Design the complete integrated system.
- Connect all purchased components correctly.
- Develop and upload the required code.
- Debug and simulate the system. Demonstrate successful operation of all three functionalities working together.

3. PixelVerse

Imagine. Prompt. Create

It is designed to test participants' ability to communicate effectively with AI through well-crafted prompts. The competition evaluates creativity, observation skills, prompt optimization, and prompt engineering techniques using AI image generation.

For this event, participants will use Google Gemini to generate images based on prompts or reference images.

Team size: Team (2-4 members)

Number of Rounds: 2

General Rules:

- Participants must use Google Gemini for image generation.
- Participants can use laptops or phone based on their convenience.
- Editing software or external AI tools are not allowed.
- Participants may modify and refine their prompts during the allotted time.
- Any attempt to cheat, misuse AI tools, or violate the event guidelines will lead to immediate disqualification & the judges' decision will be final.

Round 1 – Prompt Enhancement Challenge:

- All participants will receive the same simple one-line prompt.
- Participants are expected to improve and optimize the prompt by adding appropriate details, style, lighting, camera angle, environment, colours, artistic elements, and other creative enhancements to generate the best possible image.
- This round evaluates how effectively participants can transform a basic prompt into a professional AI prompt.

Round 1B - Image Recreation Challenge

- Participants will be shown a simple reference image.
- Their task is to carefully observe the image and write prompts that recreate it as accurately as possible using Google Gemini.
- Participants may refine and modify their prompts within the allotted time. The performance in Round 1B will determine the shortlisted participants for the Final Round.

Round 2 – Advanced Image Recreation (FINAL ROUND)

- Qualified participants will receive a highly detailed reference image.
- The image may include
Multiple objects, Complex backgrounds ,Lighting effects
,Camera angles ,Depth of field ,character expressions ,Fine textures ,Colour accuracy.
- Participants must recreate the image as closely as possible through advanced prompt engineering.
- This round focuses on professional-level prompt writing and precision.

Tips for Participants:

- Observe carefully before writing prompts.
- Include lighting, colours, composition, camera angle, style, textures, and background details whenever applicable.
- Experiment with prompt refinement.
- Use clear and descriptive language.
- Think like a Prompt Engineer, not just an AI user.

NON – TECHNICAL EVENTS

1. Mind Sprint

Think Fast. Answer Faster

Mind Sprint is an individual rapid-fire quiz designed to test participants general knowledge, quick thinking, and decision-making skills. Questions range from straightforward to moderately challenging, requiring both accuracy and speed to secure the top spot.

Rules & Regulations:

- The event consists of a single rapid-fire round.
- Participation is individual only; team participation is not permitted. Participants must use their own device and ensure a stable internet connection.
- Questions must be answered within the specified time limit.
- Each correct answer earns points; no points will be awarded for unanswered questions.
- The participant with the highest score will be declared the winner.
- In the event of a tie, the participant with the faster completion time will be ranked higher.
- Use of external resources, assistance from others, or any unfair means is strictly prohibited.
- Participants must join the quiz before the commencement of the event.
- Any form of malpractice will lead to immediate disqualification.
- The decision of the organizers will be final and binding.

Event Format:

Mode: Individual Participation only

Number of Rounds: 1

Type: Rapid-Fire Quiz

Winner Criteria:

Highest Score → Fastest Completion Time (Tie-breaker)

2. Stranger Things

Think Fast. Answer Faster

A high-stakes, mystery-themed engineering forensics competition bridging the gap between software, hardware, and mechanical logic. Teams will be plunged into the "Upside Down" of technology, where they must solve cryptic technical quizzes and visually piece together completely disassembled components to deduce the identity of real-world machines, ranging from smart IoT networks to heavy civil infrastructure

Team size: 2 to 4 members per team.

Rules & Regulations:

- The Buzzer System: All rounds operate on a live buzzer system.
- The first team to hit their buzzer gets the floor to answer.
- The "One Strike" Rule: Each team gets exactly one attempt to answer per prompt. If a team buzzes in and answers incorrectly, they are immediately locked out for the remainder of that specific question.
- Response Time: Upon buzzing in, a team has exactly 3 seconds to provide their answer. Failure to answer in time counts as an incorrect guess.
- Points: Every correct answer across all rounds is awarded 1 point. There is no negative marking for incorrect guesses.
- Tie-Breaker: In the event of a tie at the end of the competition, a sudden-death "Rapid Fire" round will commence involving a single, highly complex architectural flowchart where teams must spot a critical system flaw.

Round 1: THE UPSIDE DOWN (Cryptic Quizzes)

- A rapid-fire logic round.
- The host projects text-based quizzes describing the function, behaviour, or physical attributes of a standard engineering system (e.g., a smart lock, a mechanical piston, or an electrical transformer) without naming it.
- Teams must deduce the component based purely on its behavioural description.

Round 2: THE DEMOGORGON TEARDOWN (Visual Autopsy)

- A visual deduction challenge.
- The host projects a timed sequence of images on the screen.
- The sequence starts with obscure, completely disassembled internal parts laid out flat (like jumper wires, breadboards, or bare heating coils) and slowly adds more recognizable parts every 5 seconds.
- Teams must guess the complete, assembled system before the final image is shown.

EVENT FLOW:

Execution of Round 1:

- The host projects cryptic quizzes on the main screen one by one.
- Teams listen and read simultaneously, buzzing in the moment they deduce the engineering component or system.

Execution of Round 2:

- The room lights are dimmed for the visual deduction challenge.
- The host controls the slide deck, projecting a new piece of a disassembled machine every 5 seconds.
- Teams debate quietly and buzz in to lock in their final answer before the complete machine is revealed.

Conclusion:

The host tallies the final scores, executes the rapid-fire tie-breaker if necessary, and announces the winning forensics team.

3. Movie Mania

Lights. Camera. Challenge

Lights, camera, action! Put your movie knowledge to the test through exciting rounds of movie clips, dialogues, posters, emojis and trivia. Think fast, guess right and prove you're the ultimate movie buff!

Rules & Regulations:

- Each team must consist of 2 participants.
- One answer sheet will be provided per team.
- Teams must write only one answer for each question.
- Answers written after the allotted time will not be considered.
- Mobile phones, smart devices, smart watches, and internet access are strictly prohibited.
- Any form of malpractice or communication with other teams will lead to disqualification.
- In the event of a tie, a tie-breaker round will be conducted.
- If two or more teams secure the same total score, a Rapid-Fire Tie-Breaker will be conducted to determine the winners.
- The decision of the judges and event coordinators will be final.

Round 1 – Dialogue Detective

Identify the movie from a famous dialogue.

Rules:

A famous dialogue will be displayed or played.

Participants must identify the correct movie.

Each question has a 10-second time limit.

5 Questions × 5 Marks = 25 Marks

Round 2 – Perfect Pair

Identify the movie in which the displayed multiple character appeared in movie.

Rules:

Multiple images (Hero, Heroine, Villain) will be displayed.

Participants must identify the movie in which actors acted together.

Multiple collaborations may exist; the displayed look/costume will indicate the intended movie.

Each question has a 10-second time limit.

5 Questions × 5 Marks = 25 Marks

Round 3 – Empty Frame Challenge

Identify the movie from a scene where all characters have been removed.

Rules:

A famous movie location or scene will be shown with all people edited out.

Participants must identify the movie using the background, props, or setting.

Each question has a 10-second time limit.

5 Questions × 5 Marks = 25 Marks

Round 4 – Character Connect

Identify the character's name from the displayed image.

Rules:

An image of a movie character will be shown.

Participants must answer with the character's name, not the actor's name.

Each question has a 10-second time limit.

5 Questions × 5 Marks = 25 Marks

Event Flow:

- Each team shall consist of 2 participants.
- Every team will be provided with one answer sheet.
- The event consists of 4 rounds.
- All teams will participate in all four rounds.
- Teams must discuss and write one final answer for each question on the provided answer sheet.
- Answers must be submitted within the allotted time for each question.
- There are no elimination rounds.
- At the end of the event, the total score from all four rounds will be calculated.
- The teams with the highest overall scores will be declared the winners.

Thank you for joining us on this journey of innovation, learning and excellence.

See you at SIGNOVA '26.